Database Management System (DBMS) Lab

Lab Project 04: Bank Account Data Integrity Project

Course: DBMS Lab **Audience:** CSE Students **Keywords:** Constraints, Integrity **Env:** XAMPP, phpMyAdmin

1. SCHEMA SETUP & TABLE CREATION (SQL CONSOLE)

```sql
CREATE DATABASE IF NOT EXISTS BankingDB; USE BankingDB;
CREATE TABLE Accounts (
    account_id INT PRIMARY KEY AUTO_INCREMENT,
    account_number VARCHAR(20) UNIQUE NOT NULL,
    account_holder VARCHAR(100) NOT NULL,
    nid_number VARCHAR(17) UNIQUE NOT NULL,
    account_type VARCHAR(20) DEFAULT 'Savings',
    balance DECIMAL(12, 2) DEFAULT 500.00,
    status VARCHAR(10) DEFAULT 'Active',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_balance CHECK (balance >= 500.00),
    CONSTRAINT chk_status CHECK (status IN ('Active', 'Inactive', 'Frozen'))
);
```

2. VALID DATA INSERTION (PASSING RULES)

```sql
INSERT INTO Accounts (account_number, account_holder, nid_number, account_type, balance) VALUES ('ACC-1001', 'Anisur Rahman', '19951234567890123', 'Current', 15000.00);
INSERT INTO Accounts (account_number, account_holder, nid_number) VALUES ('ACC-1002', 'Sharmin Akter', '19989876543210987');
```

3. CONSTRAINT VERIFICATION & ERROR TESTING (RUN INDIVIDUALLY)

3.1 NOT NULL Violation

```sql
INSERT INTO Accounts (account_number, account_holder, nid_number) VALUES ('ACC-1003', NULL, '20011122334455667');
```

```
Err 1048: Column 'account_holder' cannot be null
```

3.2 UNIQUE Violation

```sql
INSERT INTO Accounts (account_number, account_holder, nid_number) VALUES ('ACC-1001', 'Kamal', '20029988776655443');
```

```
Err 1062: Duplicate entry 'ACC-1001'
```

3.3 CHECK Violation

```sql
INSERT INTO Accounts (account_number, account_holder, nid_number, balance) VALUES ('ACC-1004', 'Tariqul', '20035544332211009', 200.00);
```

```
Err 3819: Check constraint 'chk_balance'
```

4. STATE VERIFICATION & SUBMISSION REQUIREMENTS

Verification Query

```sql
SELECT * FROM Accounts;
```

Confirms blocked invalid rows and data consistency.

Submission Guidelines

- Include executable `.sql` script.
- Attach error capture screenshots.
- Zip & submit as `StudentID_DBMS_Project04.zip`.